Week 9 Homework

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In order to receive full credit problems and solutions must be clearly written and presented neatly. In writing up your solution, you should state the problem as explaining the solution. Merely giving answers to exercises will is not enough to receive credit. Your write up must show work and convince me that you understand the material. If you need additional time to complete this assignment, please email me. Starred (*) proof problems are eligible for inclusion in the proof portfolio assignment.

9.1 Exercises

9.1.1 Exercise

Solve the recurrence relation

$$a_n = \sqrt{\frac{a_{n-2}}{a_{n-1}}}$$

with initial conditions $a_0 = 8$ and $a_1 = 1/(2\sqrt{2})$ by taking the logarithm of both sides and making the substitution $b_n = \log_2 a_n$.

9.1.2 Exercise

Solve the recurrence relation

$$a_n = (a_{n-1} \cdot a_{n-2})^2$$

with initial conditions $a_0 = 1$ and $a_1 = \sqrt{2}$ by taking the logarithm of both sides and making the substitution $b_n = \log_2 a_n$.

9.1.3 Exercise

Solve the recurrence relation $T(n) = nT^2(n/2)$ with initial condition $T(1) = 6$. (Hint: Let $n = 2^k$ and then make the substitution $a_k = \log T(2^k)$.)

9.1.4 Exercise

Convert the following non-homogeneous linear recurrence relations into homogeneous linear recurrence relations by eliminating the non-homogeneous piece.

1. $a_n = 2a_{n-1} + n + 1$
2. $b_n = 3b_{n-1} - 3b_{n-2} + 2^n$

9.1.5 Exercise

Find a recurrence relation for c_n the number of ways to construct a $2 \times 2 \times n$ pillar out of $2 \times 2 \times 1$ bricks. What are the initial conditions?

9.1.6 Exercise

Find a recurrence relation for b_n the number of ways to construct a $2 \times 2 \times n$ pillar out of $2 \times 1 \times 1$ bricks. What are the initial conditions? (Though very similar to the previous problem, this version is much harder. Hint: The solution is not $b_n = 2b_{n-1} + 5b_{n-2}$, that is only part of the solution.)

9.1.7 Exercise

Find a recurrence relation for d_n the number of ways to construct a $2 \times 2 \times n$ pillar out of $2 \times 2 \times 1$ bricks and $2 \times 2 \times 2$ bricks . What are the initial conditions?

9.1.8 Exercise

Find the definition of the **complement** $\bar{G}$ of a graph G. Give three examples of a graph and it's complement.

9.1.9 Exercise

A tree with n vertices is called **graceful** if it's vertices can be labeled with the integers 1, 2 , ..., n such that the absolute values of the difference of the labels of adjacent vertices are all different. Draw and label two different graceful trees on 6 vertices.

9.1.10 Exercise

Create a graph to represent all states of the following problem where the vertices are an ordered pair of how much water is each jug (e.g (5,0) means 5 gallons in the 5 gallon jug and 0 gallons in the 3 gallon jug) and the edges represent the change levels with one pour. Problem: "You've got to defuse a bomb by placing exactly 4 gallons of water on a sensor. The problem is, you only have a 5 gallon jug and a 3 gallon jug on hand!"

9.2 Proofs

9.2.1 Proof

Prove that if there are two different paths $P_1 : x, v_1, v_2, \cdots, v_k, y$ and $P_2 : x, u_1, u_2, \cdots, u_l, y$ where at least one $v_i \neq u_j$ for some i and $j$between vertices x and y in a simple graph, then there must be a cycle in the graph.

9.2.2 Proof

Prove that if a simple graph contains an odd length closed trail $v_0 \to v_1 \to v_2 \to \cdots \to v_k$ then the graph contains an odd length cycle.

9.2.3 Proof

Prove $\frac{1}{n+1}\binom{2n}{n} = \frac{2 \cdot 6 \cdot 10 \cdots (4n-2)}{(n+1)!}$. Both of these expressions are closed form solutions to the Catalan number sequence C_n.

9.2.4 Proof

Let C_n be the Catalan number sequence, prove that $(n+1)C_n = (4n-2)C_{n-1}$.

9.2.5 Proof *

Prove the sum of the first n triangular numbers is $\frac{n(n+1)(n+2)}{6}$

9.2.6 Proof *

Prove the that the number of n-digit binary numbers with no consecutive 1's is the Fibonacci number F_{n+2}, where $F_1 = 1$ and $F_2 = 1$.

9.2.7 Proof *

Prove the following statement about the Catalan numbers C_n for all all positive integers $n \geq 1$: $C_n \geq 2^{n-1}$ (Hint: You may want to use the last part of Proof .[cproof])

9.2.8 Proof *

Prove the following statement about the Catalan numbers C_n for all all positive integers $n \geq 1$: $C_n \geq \frac{4^{n-1}}{n^2}$.